

XINZHENG, China

WMO: 570830

Lat: 34.7064N

Lon: 113.6628E

Elev: 111

StdP: 100.00

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.1	(c) -4.9	(d) -22.2	(e) 0.5	(f) 2.7	(g) -19.5	(h) 0.7	(i) 2.8	(j) 10.4	(k) 3.1	(l) 8.7	(m) 2.9	(n) 1.6	(o) 270	(p) 0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.2	(c) 35.7	(d) 23.5	(e) 34.2	(f) 23.8	(g) 33.0	(h) 23.7	(i) 28.0	(j) 31.8	(k) 27.2	(l) 31.0	(m) 26.4	(n) 30.1	(o) 3.2	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 27.1	(b) 23.1	(c) 30.6	(d) 26.2	(e) 21.9	(f) 29.8	(g) 25.4	(h) 20.8	(i) 29.0	(j) 91.2	(k) 31.9	(l) 87.5	(m) 31.2	(n) 83.8	(o) 30.1	(p) 30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 6.8	(c) 5.7	(e) -9.2	(f) 38.7	(g) 1.8	(h) 1.4	(i) -10.5	(j) 39.7	(k) -11.5	(l) 40.5	(m) -12.5	(n) 41.3	(o) -13.8	(p) 42.3		
(a) 8.2	(b) 6.8	(c) 5.7	(e) -10.3	(f) 29.4	(g) 1.7	(h) 0.9	(i) -11.5	(j) 30.1	(k) -12.5	(l) 30.6	(m) -13.4	(n) 31.2	(o) -14.6	(p) 31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	1.1	4.3	10.2	16.4	21.9	26.3	27.7	26.3	21.8	16.4	9.2	3.3		
	DBStd	9.81	3.21	3.83	4.27	4.02	3.25	2.76	2.50	2.75	3.10	3.35	4.12	3.20		
	HDD10.0	766	276	163	51	4	0	0	0	0	0	2	62	209		
	HDD18.3	2099	534	393	254	83	8	0	0	0	7	78	275	467		
	CDD10.0	2759	1	3	56	195	368	490	548	507	354	199	37	1		
	CDD18.3	1049	0	0	2	24	117	240	290	248	111	17	0	0		
	CDH23.3	10744	0	0	24	241	1116	2663	3247	2473	824	151	6	0		
	CDH26.7	4471	0	0	4	63	413	1264	1463	980	256	27	0	0		
(14) Wind	WSAvg	2.4	2.5	2.6	2.9	2.8	2.7	2.4	2.2	2.2	2.0	2.1	2.4	2.6		
(15) Precipitation	PrecAvg	636	9	13	24	43	56	61	146	124	85	42	25	9		
	PrecMax	1043	54	65	96	194	227	231	369	270	253	156	117	55		
	PrecMin	354	0	0	0	2	0	2	34	32	0	0	0	0		
	PrecStd	156	11	12	21	37	47	51	83	64	63	37	25	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	18.2	26.2	31.3	35.9	37.6	37.2	36.1	33.9	29.7	24.1	16.3		
		MCWB	5.6	9.3	13.8	19.1	18.1	21.4	24.8	26.5	22.7	17.8	13.3	6.7		
	2%	DB	11.6	15.5	23.0	28.3	32.8	35.7	35.3	34.0	31.1	26.9	20.8	13.4		
		MCWB	4.1	7.5	13.3	17.1	18.7	21.6	25.6	26.0	22.0	15.9	11.9	5.3		
	5%	DB	9.1	13.2	20.4	26.1	30.8	34.2	34.1	32.5	29.5	24.9	18.2	11.2		
		MCWB	2.7	6.2	11.8	16.5	18.7	21.6	25.8	25.5	20.7	15.7	10.8	4.3		
	10%	DB	7.1	11.1	18.0	23.9	28.9	32.8	32.8	31.2	27.9	22.9	16.1	9.1		
		MCWB	1.6	5.2	10.4	15.4	18.4	21.4	25.7	24.9	20.5	15.1	9.4	3.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	11.3	17.1	21.9	23.6	27.0	29.5	29.0	26.6	20.0	15.5	8.6		
		MCDB	12.8	16.1	22.9	27.4	28.9	32.0	33.5	33.4	30.4	23.7	20.1	13.6		
	2%	WB	4.6	8.9	14.6	19.7	22.1	25.6	28.3	27.9	24.6	18.7	13.2	6.5		
		MCDB	10.1	13.3	20.7	25.3	27.8	31.1	32.3	31.3	28.1	23.5	18.5	10.6		
	5%	WB	3.3	7.3	12.8	18.1	20.9	24.5	27.6	27.1	23.1	17.6	11.9	5.3		
		MCDB	8.0	11.5	18.9	24.0	26.8	30.0	31.5	30.5	26.8	21.8	16.5	9.7		
	10%	WB	2.2	5.9	11.2	16.7	19.9	23.5	26.8	26.3	21.8	16.5	10.6	4.2		
		MCDB	6.2	10.2	16.8	21.9	25.8	29.1	30.7	29.6	25.3	20.6	14.8	8.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.6	10.3	10.9	11.0	10.6	8.2	7.9	9.2	10.0	9.6	8.8		
		MCDBR	13.1	13.8	14.7	14.6	14.6	13.6	10.5	9.8	12.2	13.8	13.9	13.4		
	5% WB	MCWBR	7.5	7.6	7.3	6.8	5.4	4.5	4.0	3.9	4.6	5.6	7.0	7.4		
		MCWBR	11.6	12.2	13.5	12.6	11.3	10.6	8.6	8.3	9.5	10.4	12.0	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.579	0.651	0.655	0.663	0.677	0.795	0.778	0.779	0.727	0.712	0.632	0.566		
	taud		1.645	1.547	1.563	1.580	1.601	1.434	1.510	1.490	1.544	1.519	1.605	1.682		
	Ebn at Noon		574	586	640	668	666	590	597	583	582	533	525	552		
	Edh at Noon		200	245	260	267	265	313	288	289	262	248	207	182		
(44) All-Sky Solar Radiation	RadAvg		2.27	2.77	3.99	4.77	5.22	5.07	4.53	4.28	3.61	2.98	2.42	2.18		
	RadStd		0.26	0.49	0.39	0.40	0.44	0.37	0.44	0.57	0.46	0.37	0.40	0.32		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+84	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	N/A

Nomenclature: See separate page